

Technical Report:

Government Arts And Science College Manur, Tirunelveli.

College code: msu142

Department: [B.SC](#) Computer science

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Implement Client Script & UI Policy (Incident)

1. Executive Summary

Problem Statement

Incident records in ServiceNow often require consistent and accurate data entry to facilitate efficient triage, routing, and resolution. Relying solely on manual checks or end-user awareness frequently leads to incomplete, inconsistent, or invalid data submissions. This degrades reporting accuracy, compromises SLA compliance, and impacts overall service quality. Implementing client-side controls directly at the user interface level enforces conditional field behavior and validation before records are committed to the database.

Objective

This project demonstrates the implementation of ServiceNow client-side controls—specifically **UI Policies** and **Client Scripts**—to enforce data integrity on Incident records. The primary goal is to dynamically control field visibility, enforce mandatory field requirements, auto-populate dependent field values, and prevent form submission or list-level edits when business conditions are not satisfied.

Core Technical Skills

- Incident Management
- UI Policies & UI Policy Actions
- Client Scripts (onChange, onSubmit, onCellEdit)
- Form Validation & User Interface Controls

2. Configuration & Implementation

Task 1: Create UI Policy on Incident

The **High Impact Control** UI Policy dynamically triggers business rules when an Incident's impact is classified as high.

- **Navigation:** System UI \rightarrow UI Policies
- **Name:** High Impact Control
- **Table:** Incident (incident)
- **Active:** true
- **Condition:** Impact is 1 - High
- **Reverse if false:** true (automatically reverts applied field modifications when conditions are no longer met)
- **UI Policy Action:** Set Assignment group as mandatory

Short description	Table	Conditions	Reverse if false	On load	Updated	Order
High Impact Control	Incident [incident]	impact=1^EQ	true	true	2026-09-21 23:47:04	100
Hide for Daily Rotations	Roster [cmn_rota_roster]	rotation_interval_type=daily^EQ	true	true	2026-06-13 03:10:31	100
Weekly Rotations Not All Day	Roster [cmn_rota_roster]	rotation_interval_type=weekly^rotation_a...	false	true	2026-06-13 03:10:31	100
Weekly Rotations All Day	Roster [cmn_rota_roster]	rotation_interval_type=weekly^rotation_a...	false	true	2026-06-13 03:10:31	100
No Reminders	Roster [cmn_rota_roster]	send_reminders=false^EQ	true	true	2026-06-13 03:10:31	100
Show pricing implications checkbox	Question [question]	type=8^ORtype=21^EQ	true	true	2026-06-13 02:56:32	100
Show price if checked	Question [question]	type=7^EQ	true	true	2026-06-13 02:56:32	100
Duration only applies to request type categories	Dynamic category [sc_category_top_n]	type=sc_req_item^EQ	true	true	2026-06-13 02:56:32	100
Show Multiple Choices and Choice price	Item Option [sc_cat_item_option]	type=multiple_choice^ORtype=numeric_scal...	true	true	2026-06-13 02:56:32	100

UI Policy - High Impact Control

UI policies change fields on a form based on a set of conditions. Use UI policies to show or hide fields, or to make fields mandatory or read-only based on these conditions. A UI policy specifies one or more actions to take when the policy is triggered. First, create the policy. Then add as many actions as necessary. UI policy actions are applied only if all of the following conditions are met:

- The UI Policy is **Active**
- The items in the **Conditions** field evaluate to true
- The field specified in the UI policy action is present on the specified form

[More info](#)

Table: Incident [incident] Application: Global Active: ☒

* Short description: High Impact Control

When to Apply

Conditions: Add Filter Condition Add OR Clause

Impact is 1 - High AND OR

Update Delete

Related Links

UI Policy Action - assignment_group

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More info](#)

UI policy: High Impact Control Application: Global

Table: Incident [incident] Mandatory: True

* Field name: Assignment group Visible: Leave alone

Read only: False

Clear the field value: ☐

Update Delete

Task 2: Create UI Policy Action – Urgency Field

To lock field modification when high-impact conditions exist, a UI Policy Action is configured on the Urgency field.

- **Parent Policy:** High Impact Control
- **Field Name:** Urgency (urgency)
- **Read-only:** true

- **Visible:** Unchanged (remains visible on the form)

The screenshot shows the ServiceNow 'UI Policy Action - urgency' configuration page. The page is titled 'UI Policy Action - urgency' and includes a search bar and navigation links. The main configuration area is divided into two sections: 'UI policy' and 'Application'.

UI policy section:

- UI policy: High Impact Control
- Table: Incident [incident]
- * Field name: Urgency

Application section:

- Application: Global
- Mandatory: Leave alone
- Visible: Leave alone
- Read only: True
- Clear the field value: ☐

At the bottom of the configuration area, there are 'Update' and 'Delete' buttons. Below the configuration area is a large empty space with a 'Refresh' button in the bottom right corner.

Task 3: Create onChange Client Script

An onChange Client Script automatically sets the Urgency field to High whenever a user updates the Impact field to High.

- **Navigation:** System UI \rightarrow Client Scripts
- **Name:** Auto set urgency for high impact
- **Table:** Incident (incident)
- **Type:** onChange
- **Field Name:** Impact (impact)
- **Active:** true

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (newValue == '1') {
        g_form.setValue('urgency', '1');
        g_form.addInfoMessage('Urgency set to High for High impact incident.');
```

ServiceNow Developers

Client Scripts | ServiceNow

Ask Gemini

dev311746.service-now.com/now/nav/ui/classic/params/target/sys_script_client_list.do%3Fsysparm_userpref_module%3D09b4f089c0a800050098a7faefe5d732%26s...

servicenow

All Favorites History Workspaces

Client Scripts

Search

Actions on selected rows...

New

All > Class = Client Script

Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [sn_templated_snip_channel]	Templated Snippets		onChange	2020-09-22 17:06:27
Show overlap field based on data labels	true	Widgets [pa_widgets]	Global		onChange	2021-04-23 02:59:51
Set protocol based on auth type	false	HTTP(s) Connection [http_connection]	Global		onChange	2017-06-02 15:49:24
Set default values for OIDC PKCE flow	true	OAuth OIDC Entity [oauth_oidc_entity]	Global		onChange	2020-02-28 00:59:58
Make notification read only	true	Additional Notification Recipients [sys_notification_recipient]	Global		onLoad	2021-02-05 22:42:13
Deprecated - SN - Make serial number man	false	Asset [alm_asset]	Global		onChange	2025-03-06 00:42:54
Validate StatefulSet Name	true	MID K8s Deployment Request [mid_k8s_deployment]	Global		onChange	2022-10-09 23:18:07
Set Use current record if delete	true	Action item [sys_sg_write_back_action_item]	Global		onChange	2019-04-14 07:30:02

1 to 20 of 2,432

32°C Light rain

Search

ENG IN

03:15 PM 22-09-2026

ServiceNow Developers

Auto set urgency for high impact

Ask Gemini

dev311746.service-now.com/now/nav/ui/classic/params/target/sys_script_client.do%3Fsys_id%3D94a1c1a4836f4b10d46ec265eeaad3c9%26sysparm_record_target%...

servicenow

All Favorites History

Client Script - Auto set urgency for h...

Search

Update Delete

Client Script
Auto set urgency for high impact

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Auto set urgency for high impact

Application: Global

Table: Incident [incident]

Active: ☒

UI Type: Desktop

Inherited: ☐

Type: onChange

Global: ☒

Field name: Impact

Description:

Messages:

Script

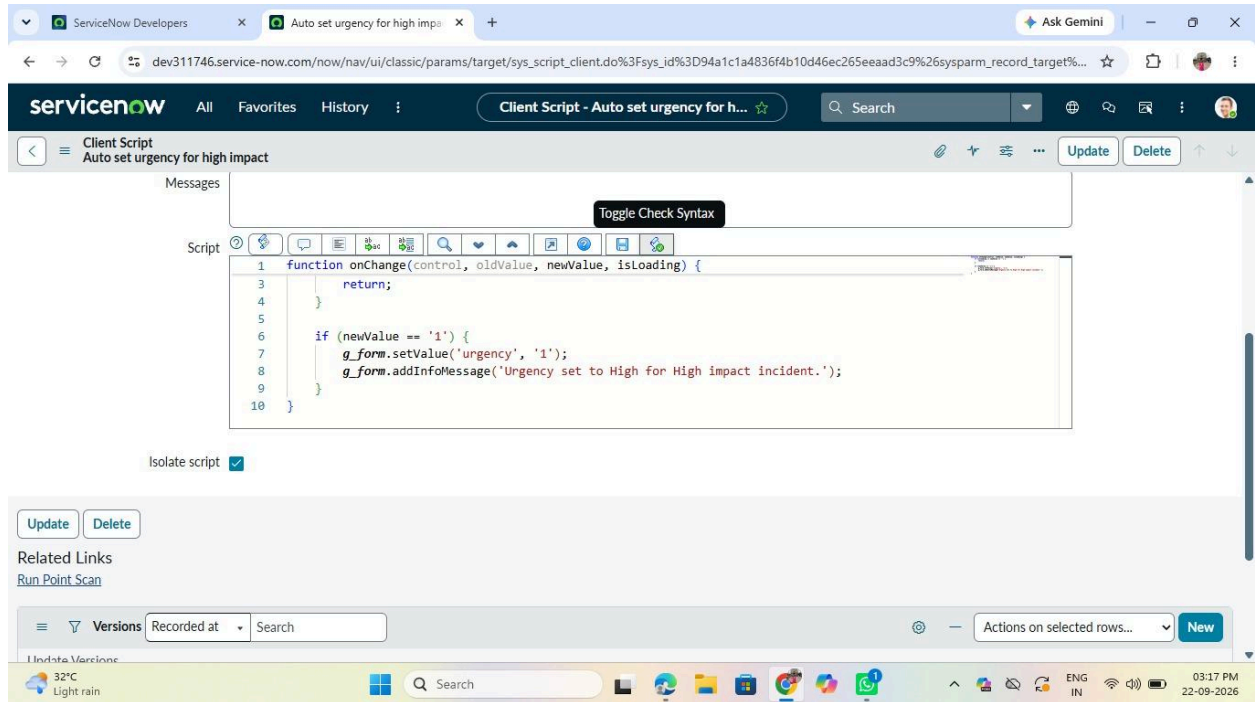
```
1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || newValue == '') {
3     return;
4   }
5 }
```

32°C Light rain

Search

ENG IN

03:17 PM 22-09-2026



Task 4: Create onSubmit Client Script (Save Validation)

An onSubmit Client Script prevents the form from submitting if Assigned To is missing on High Impact incidents.

- **Navigation:** System UI \rightarrow Client Scripts
- **Name:** Prevent save if Assigned To missing
- **Table:** Incident (incident)
- **Type:** onSubmit
- **Active:** true

```

function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
g_form.getValue('assigned_to') == '') {
        g_form.showErrorBox(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
        return false;
    }
    return true;
}

```

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Prevent save if Assigned To missing Application: Global

Table: Incident [incident] Active: ☒

UI Type: Desktop Inherited: ☐

Type: onSubmit Global: ☒

Description:

Messages:

Script

```

1 function onSubmit() {
2   if (g_form.getValue('impact') == '1' &&
3       g_form.getValue('assigned_to') == '') {
4
5     g_form.showErrorMessage(
6       'assigned_to',
7       'Assigned To is mandatory for High impact incidents.'
8     );
9     return false;
10
11   }
12   return true;
13 }
14

```

Isolate script: ☒

Update Delete

Related Links

[Run Point Scan](#)

Versions Recorded at Search

Actions on selected rows... New

Task 5: Create onCellEdit Client Script

An onCellEdit Client Script blocks users from modifying the State field directly from the Incident list view, mandating that changes occur through the form view.

- **Navigation:** System UI \rightharpoonarrow Client Scripts
- **Name:** Prevent state change via list edit

- **Table:** Incident (incident)
- **Type:** onCellEdit
- **Field Name:** State (state)
- **Active:** true

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
    alert('State cannot be updated using list editing. Please open the Incident.');
```

```
    callback(false);
}
```

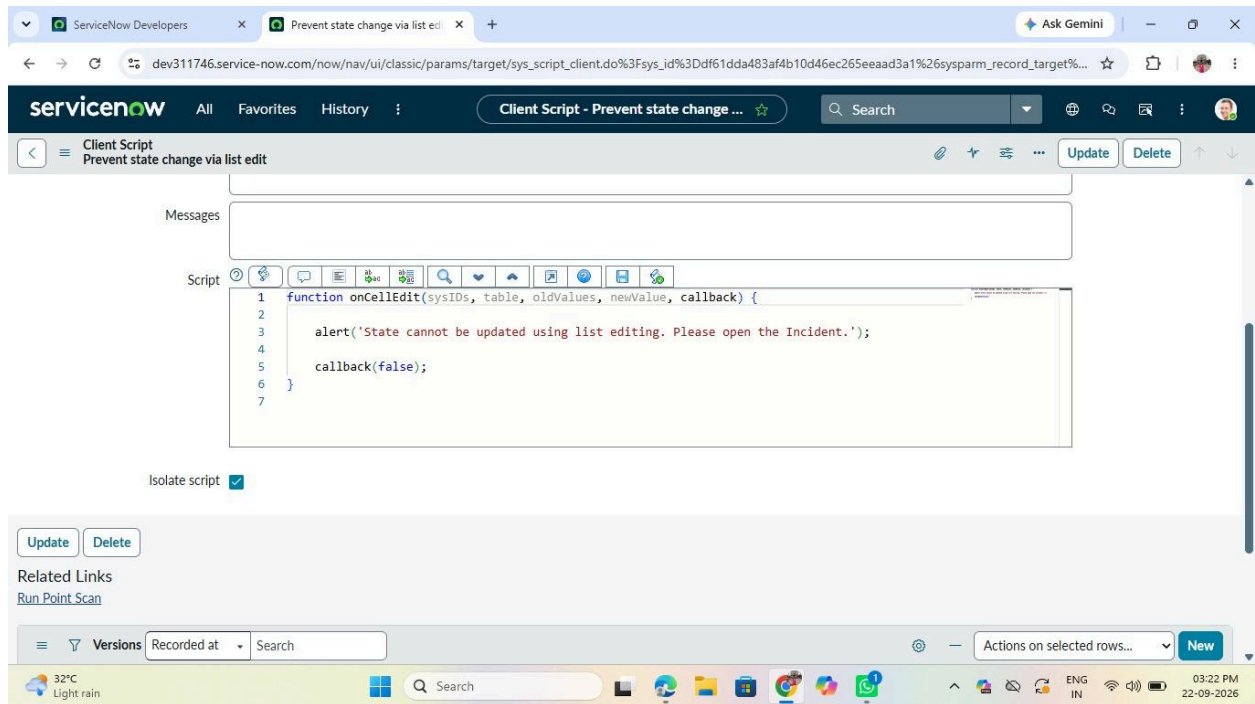
The screenshot shows the ServiceNow interface for configuring a Client Script. The browser tabs include 'ServiceNow Developers' and 'Prevent state change via list edit'. The URL is a long alphanumeric string. The page title is 'Client Script - Prevent state change ...'. A search bar is visible. Below the title, there's a notification about client-scripts running in strict mode. The configuration form includes the following fields:

- Name:** Prevent state change via list edit
- Table:** Incident [incident]
- UI Type:** Desktop
- Type:** onCellEdit
- Field name:** State
- Application:** Global
- Active:** ☒
- Inherited:** ☐
- Global:** ☒
- Description:** (empty text area)
- Messages:** (empty text area)
- Script:**

```
1 function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
2
3     alert('State cannot be updated using list editing. Please open the Incident.');
```

```
4
5     callback(false);
6 }
```

The bottom of the screen shows a Windows taskbar with a search bar, various application icons, and system status information: 32°C, Light rain, ENG IN, and 03:22 PM 22-09-2026.



3. Testing & Verification

Test Case	Procedure	Expected Result	Pass/Fail Criteria
Activity 1: Mandatory Enforcement	Open new Incident, set Impact = High, leave Assigned To empty, click Submit .	Form submission is blocked; field error box appears under Assigned To.	Record does not save; error banner displays.
Activity 2: Successful Save	Populate Assigned To with a valid user on a High Impact Incident, click Submit .	Form submits successfully; Urgency auto-sets to High; Urgency is read-only.	Record saves without errors.
Activity 3: Reverse Condition Test	Open an Incident where Impact = High, change Impact to Medium, save.	Urgency becomes editable; Assigned To requirement relaxes per policy reverse settings.	Dynamic reversal of UI rules triggers correctly.
Activity 4: List Edit Blocking	Navigate to Incident list view (Incident → All), double-click State cell to edit.	Alert box displays warning message; edit is cancelled via callback(false).	Cell edit reverts; state value remains unchanged.
Activity 5: Form-Based Update	Open Incident record form, update State field, click Update .	Form updates successfully.	Record updates without list-edit restrictions.

ServiceNow Developers

Incidents | ServiceNow

+

Ask Gemini

dev311746.service-now.com/now/nav/ui/classic/params/target/incident_list.do%3Factive%3Dtrue%26sysparm_query%3Dactive%253Dtrue%255EEQ

servicenow

All Favorites History Workspaces

Incidents

Search

Incidents

Number

Search

Actions on selected rows...

New

All > Active = true

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010007	2026-09-22 01:37:02	testing UI policy	Adela Cervantsz	1 - Critical	In Progress	Inquiry / Help	Software	Beth Anglin	2026-09-22 01:39:22	admin
INC0010003	2026-09-22 01:09:35	Hii	Abel Tuter	1 - Critical	In Progress	Inquiry / Help	CAB Approval	Bernard Laboy	2026-09-22 01:11:39	admin
INC0010002	2026-09-22 01:00:32	practise	Abel Tuter	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-09-22 01:49:10	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin

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32°C Light rain

Search

ENG IN

03:39 PM 22-09-2026

ServiceNow Developers

Create INC0010007 | Incident

+

Ask Gemini

dev311746.service-now.com/now/nav/ui/classic/params/target/incident.do%3Fsys_id%3DINC0010007%26sysparm_query%3Dsys_id%3DINC0010007%26sysparm_target%3Dincident%26sysparm_check%3Dtrue

servicenow

All Favorites History Admin

Incident - Create INC0010007

Search

Incident

New record

Submit

Resolve

Number

INC0010007

Channel

-- None --

* Caller

Adela Cervantsz

State

New

Category

Inquiry / Help

Impact

1 - High

Subcategory

-- None --

Urgency

1 - High

Service

Priority

1 - Critical

Service offering

* Assignment group

Software

Assigned to

Configuration item

* Short description

testing UI policy

Description

Assigned To is mandatory for High impact incidents.

Related Search Results

Related Search

testing UI policy

Knowledge & Catalog (All)

32°C Light rain

Search

ENG IN

03:28 PM 22-09-2026

ServiceNow Developers x Create INC0010007 | Incident | x + Ask Gemini

dev311746.service-now.com/now/nav/ui/classic/params/target/incident.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dincident%26sysparm_chec...

servicenow All Favorites History Admin Incident - Create INC0010007 Search Submit Resolve

Incident New record

Urgency set to High for High impact incident.

Number: INC0010007
* Caller: Adela Cervantsz
Category: Inquiry / Help
Subcategory: -- None --
Service:
Service offering:
Configuration item:
* Short description: testing UI policy
Description:
Channel: -- None --
State: New
Impact: 1 - High
Urgency: 1 - High
Priority: 1 - Critical
* Assignment group: Software
Assigned to: Beth Anglin

Related Search Results

Related Search: testing UI policy Knowledge & Catalog (All)

32°C Light rain 03:29 PM 22-09-2026

ServiceNow Developers x INC0010007 | Incident | Service | x + Ask Gemini

dev311746.service-now.com/now/nav/ui/classic/params/target/incident.do%3Fsys_id%3Df3c8112883238b10d46ec265eeaad38e%26sysparm_record_target%3...

servicenow All Favorites History Admin Incident - INC0010007 Search Discuss Follow Update Resolve Delete

Incident INC0010007

Number: INC0010007
* Caller: Adela Cervantsz
Category: Inquiry / Help
Subcategory: -- None --
Service:
Service offering:
Configuration item:
* Short description: testing UI policy
Description:
Channel: -- None --
State: In Progress
Impact: 2 - Medium
Urgency: 1 - High
Priority: 2 - High
Assignment group: Software
Assigned to:
Notes
Related Records
Resolution Information
Watch list
Work notes list

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ServiceNow Developers Incidents | ServiceNow

dev311746.service-now.com/now/nav/ui/classic/params/target/incident_list.do%3Fsysparm_userpref_module%3Db55fbec4c0a800090088e83d7ff500de%26sys...

servicenow All Favorites History Workspaces Incidents Search

Incidents Number Search Actions on selected rows... New

All > Active = true

	Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
	Search	Search	Search	Search	Search	Search	Search	Search	Search	Search	Search
	INC0010007	2026-09-22 01:37:02	testing UI policy	Adela Cervantsz	1 - Critical	In Progress	Inquiry / Help	Software	Beth Anglin	2026-09-22 01:39:22	admin
	INC0010003	2026-09-22 01:09:35	Hii	Abel Tuter	1 - Critical	In Progress	Inquiry / Help	CAB Approval	Bernard Laboy	2026-09-22 01:11:39	admin
☐	INC0010002	2026-09-22 01:00:32	practise	Abel Tuter	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2026-09-22 01:02:52	admin
	INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
	INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
	INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
	INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin

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32°C Light rain Search 03:35 PM 22-09-2026

ServiceNow Developers Incidents | ServiceNow

dev311746.service-now.com/now/nav/ui/classic/params/target/incident_list.do%3Fsysparm_userpref_module%3Db55fbec4c0a800090088e83d7ff500de%26sys...

servicenow All Favorites History Workspaces Incidents Search

Incidents Number Search Actions on selected rows... New

All > Active = true

dev311746.service-now.com says
State cannot be updated using list editing. Please open the Incident.
OK

	Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
	INC0010007	2026-09-22 01:37:02	testing UI policy	Adela Cervantsz	1 - Critical	In Progress	Inquiry / Help	Software	Beth Anglin	2026-09-22 01:39:22	admin
	INC0010003	2026-09-22 01:09:35	Hii	Abel Tuter	1 - Critical	In Progress	Inquiry / Help	CAB Approval	Bernard Laboy	2026-09-22 01:11:39	admin
	INC0010002	2026-09-22 01:00:32	practise	Abel Tuter	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-09-22 01:02:52	admin
	INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
	INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
	INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
	INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin

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32°C Light rain Search 03:36 PM 22-09-2026

The top screenshot displays the ServiceNow 'Incident - INC0010002' form. A context menu is open on the left, showing options like 'Save', 'Copy Incident', 'Create Incident Task', etc. The form fields include 'Number' (INC0010002), 'Channel' (None), 'State' (In Progress), 'Impact' (3 - Low), 'Urgency' (3 - Low), and 'Priority' (5 - Planning). The bottom screenshot shows the 'Incidents' list view with a table of incident records.

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010007	2026-09-22 01:37:02	testing UI policy	Adela Cervantsz	1 - Critical	In Progress	Inquiry / Help	Software	Beth Anglin	2026-09-22 01:39:22	admin
INC0010003	2026-09-22 01:09:35	Hii	Abel Tuter	1 - Critical	In Progress	Inquiry / Help	CAB Approval	Bernard Laboy	2026-09-22 01:11:39	admin
INC0010002	2026-09-22 01:00:32	practise	Abel Tuter	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-09-22 01:49:10	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin

4. Technical Conclusion

The implementation demonstrates how ServiceNow **UI Policies** and **Client Scripts** synergize to maintain database integrity, automate key fields, and control user actions at both the form and list levels. By pairing declarative UI Policies with targeted JavaScript client scripts, essential fields like Assigned To and Urgency adapt dynamically based on Impact values. This setup

reduces manual data entry errors, improves process compliance, and preserves high service management standards.